

Nomenclature of straight-chain organic compounds

Learning outcomes

At the end of this section you should be able to apply IUPAC nomenclature to saturated or mono-unsaturated compounds that have up to six carbon atoms in the parent chain and contain one type of the following functional groups: halogeno, hydroxy (hydroxyl), carbonyl, carboxyl.

What's in a name? Consider the compound named oxidane. It is a colourless compound that has no smell and no taste, yet it plays such an important part in our everyday lives. Plants cannot grow without it and humans cannot survive for more than 3 days without it. Have you guessed what it is yet? Check your answer below when you are ready.

Oxidane is the systematic IUPAC name for water, H_2O . Of course, it is rarely used as it is commonly called water.

International Mindedness

The IUPAC (International Union of Pure and Applied Chemistry) is an international organisation that aims to establish a consistent system of nomenclature (naming) for organic compounds.

In this section, you will learn how to name organic compounds using the IUPAC nomenclature rules. In this system, compounds are named using the following:

- a stem name that depends on the number of carbon atoms in the longest continuous carbon chain, known as the parent chain
- a suffix that differs according to the functional group present in the compound
- a prefix that is used to identify the position of any substituent groups.

This is summarised in **Figure 1**.



Figure 1. Organic compounds are named using a substituent, parent chain and suffix.

 More information for figure 1

The diagram illustrates the components involved in naming organic compounds. It is composed of three colored blocks arranged horizontally. Each block represents a different naming component: the left block is labeled 'Substituents,' the middle block is labeled 'Parent chain,' and the right block is labeled 'Suffix.' These components are connected to illustrate the sequence used in naming organic compounds. Each block signifies a step in the naming process, with the 'Parent chain' as the central element determining the main structure, flanked by 'Substituents' and 'Suffix.' The layout explains how these elements are combined to form the complete name of an organic compound.

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Identifying the parent chain will determine the stem name. **Table 1** shows the stem names used for up to six atoms in the parent chain. It is important to remember these as they form the basis of naming all the organic compounds in this section.

Table 1. Stem names, classes and suffixes.

Number of carbon atoms in the parent chain	Stem name	Class	Suffix
1	meth-	Alkane	-ane
2	eth-	Alkene	-ene
3	prop-	Alkyne	-yne
4	but-	Alcohol	-ol
5	pent-	Aldehyde	-al
6	hex-	Ketone	-one
		Carboxylic acid	-oic

Alkanes

To name straight-chain alkanes, first count the number of carbon atoms in the parent chain which gives the stem name, then add the suffix -ane. So, from one to six carbon atoms in the parent chain we have methane, ethane, propane, butane, pentane and hexane. **Figure 2** shows the first four members of the alkane homologous series, methane to butane.

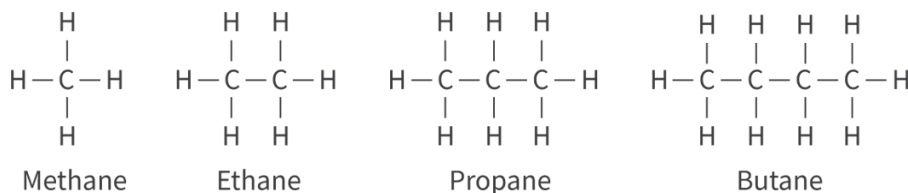


Figure 2. The first four members of the alkane homologous series methane to butane.

 More information for figure 2

The image depicts the structural formulas of the first four alkanes: methane, ethane, propane, and butane.

1. **Methane (CH₄):** Drawn as a central carbon atom bonded to four hydrogen atoms, forming a tetrahedral shape.
2. **Ethane (C₂H₆):** Consists of two carbon atoms connected by a single bond. Each carbon is further bonded to three hydrogen atoms, making a total of six hydrogen atoms.
3. **Propane (C₃H₈):** Composed of three carbon atoms connected in a straight chain with single bonds. The terminal carbon atoms are each bonded to three hydrogen atoms, while the middle carbon is bonded to two hydrogen atoms.
4. **Butane (C₄H₁₀):** Comprises four carbon atoms in a single chain, with terminal carbon atoms bonded to three hydrogens each, and the two internal carbon atoms bonded to two hydrogens each.

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Alkenes

The alkenes are named in a similar way to the alkanes, except that we need to consider the position of the C=C bond in the molecule. The parent chain gives the stem name and the suffix added is -ene (note that the parent chain must contain the C=C bond). The first two members of the homologous series have two and three carbon atoms in the parent chain, respectively. So, with two carbon atoms in the parent chain, we have ethene and with three carbon atoms we have propene (**Figure 3**).

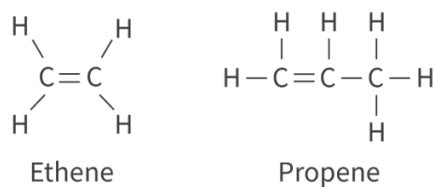


Figure 3. The first two members of the alkene homologous series: ethene and propene with two and three carbon atoms in the parent chains, respectively.

More information for figure 3

The image shows the structural formulas of the first two members of the alkene homologous series: ethene and propene. On the left, the structure of ethene is depicted with two carbon atoms connected by a double bond (C=C) and each carbon atom bonded to two hydrogen atoms (H). On the right, the structure of propene is shown with three carbon atoms in a chain. The first two carbons are connected by a double bond (C=C), and the third carbon is single-bonded (C-C) to the second. The carbon atoms each form additional single bonds with hydrogen atoms. Ethene is labeled below its structure, and propene is labeled under its respective structure.

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Once there are four carbon atoms in the parent chain, a number is used to show the position of the C=C bond. When assigning a number to the double bond, always count from the end of the molecule that gives the C=C bond the lowest possible number. So, the compounds shown in **Figure 4** are named but-1-ene and but-2-ene, respectively.

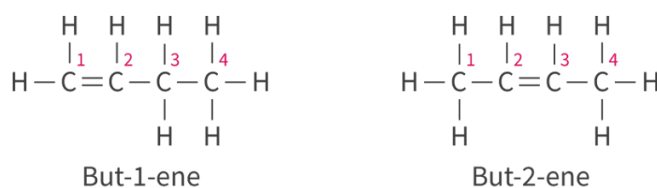


Figure 4. But-1-ene and but-2-ene are named with a number indicating the position of the C=C bond.

More information for figure 4

The image consists of two chemical diagrams labeled "But-1-ene" and "But-2-ene." Each diagram depicts a linear structure of carbon and hydrogen atoms, indicating the positioning of the carbon-carbon double bond (C=C bond).
 For But-1-ene on the left, the double bond is between the first and second carbon atoms, as numbered 1 and 2, with the chain following this order: H-C (1) = C (2)-C (3)-C (4)-H. Each carbon atom is bonded to enough hydrogen atoms to complete the four bonds around carbon.
 For But-2-ene on the right, the double bond is between the second and third carbon atoms, as numbered 2 and 3, following the sequence: H-C (1)-C (2) = C (3)-C (4)-H. Again, each carbon atom is bonded to the necessary hydrogen atoms.
 The hydrogen atoms are depicted bonded to carbon atoms all around the

chain, in both structures, maintaining the standard valency of carbon as four. This layout is typical in structural diagrams, representing molecular geometry and the precise location of the functional groups or bonds within the molecules.

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Figure 5 shows the incorrect numbering of the carbon atoms in the molecule. It is important to remember that, regardless of which end you start numbering from (left to right or right to left), the position of the functional group must always be assigned the lowest possible number. So, but-3-ene and but-4-ene do not exist as this would not give the lowest possible number for the C=C bond.

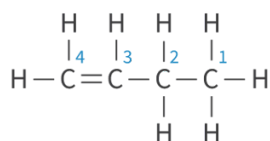


Figure 5. The incorrect numbering of the carbon atoms in but-1-ene.

 More information for figure 5

The image is a structural diagram of but-1-ene, a hydrocarbon molecule with four carbon atoms (C) and six hydrogen atoms (H). The incorrect numbering of carbon atoms is shown in blue.

1. The chain is presented in a linear format, with single and double bonds indicated.
2. The carbons are numbered from right to left as 1, 2, 3, and 4.
3. A double bond exists between the carbons labeled 3 and 4, whereas the correct IUPAC naming should have the double bond between 1 and 2 (from the left).
4. Each carbon atom has hydrogen atoms (H) attached. Specifically, carbon 1 is bonded to two hydrogens, carbons 2 and 3 each have one hydrogen, and carbon 4 has three hydrogen atoms bonded.

This illustrates the incorrect numbering method, highlighting the need for lowest possible numbering for functional groups.

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Alkynes

The alkynes are named in a similar way to the alkenes in that a number is used once the parent chain contains four or more carbon atoms. The stem name is determined by the number of carbon atoms in the parent chain and the suffix used

is -yne. The first two members of the series are ethyne and propyne (**Figure 6**).

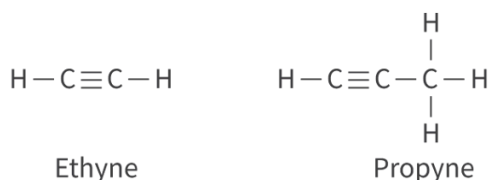


Figure 6. Ethyne (left) and propyne (right), the first two members of the alkyne homologous series.

 More information for figure 6

The image shows structural formulas of two alkyne molecules. On the left is ethyne, depicted as $\text{H}-\text{C}\equiv\text{C}-\text{H}$, with a linear arrangement representing the carbon-carbon triple bond between the two carbon atoms and a hydrogen atom bonded to each carbon. On the right is propyne, shown as $\text{H}-\text{C}\equiv\text{C}-\text{CH}_3$. Here, the triple bond connects the first two carbon atoms, and the third carbon forms part of a methyl group (CH_3) bonded to the second carbon. Both structures highlight the characteristic carbon-carbon triple bond of alkynes.

[Generated by AI]

Once there are four carbon atoms in the parent chain a number is used to show the position of the $\text{C}\equiv\text{C}$ bond. Try to name the two alkynes shown in **Figure 7**. Remember to use the lowest number possible for the location of the $\text{C}\equiv\text{C}$ bond. Check your answer when you have named them.

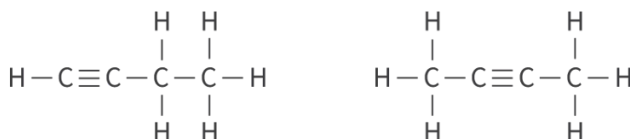


Figure 7. Try to name the two alkynes with four carbon atoms in the parent chain — remember to use the lowest possible number for the position of the $\text{C}\equiv\text{C}$ bond.

 More information for figure 7

The image shows two structural formulas of alkynes. Both molecules consist of four carbon atoms. In the first alkyne, a triple bond ($\text{C}\equiv\text{C}$) is present between the first and second carbon atoms. The first carbon is bonded to one hydrogen atom, and the fourth carbon is also bonded to three hydrogen atoms. In the second alkyne, the triple bond is between the second and third carbon atoms. The first and the fourth carbon are bonded to three hydrogen atoms each. Both structures emphasize the linear arrangement typical of alkynes and differ only in the position of the triple bond.

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The alkyne on the left is named but-1-yne and on the right but-2-yne. But-1-yne has the $\text{C}\equiv\text{C}$ bond between carbon number 1 and carbon number 2. But-2-yne has the $\text{C}\equiv\text{C}$ bond between carbon number 2 and carbon number 3. But-3-yne and but-4-yne do not exist because a lower number can be used to indicate the position of the $\text{C}\equiv\text{C}$ bond.

Alcohols

The alcohols are named by removing the 'e' from the parent alkane and replacing it with the suffix $-\text{ol}$. With one carbon atom and two carbon atoms in the parent chain, respectively, we have methanol and ethanol (**Figure 8**).

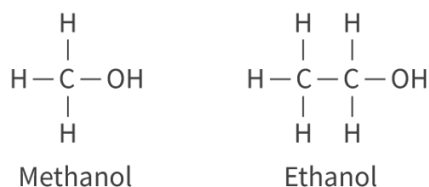


Figure 8. Methanol (left) and ethanol (right), alcohols with one and two carbon atoms in the parent chain, respectively.

 More information for figure 8

The image displays two structural formulas for alcohols. On the left is methanol, which has one carbon atom. It is shown as a central carbon (C) bonded to three hydrogen atoms (H) and one hydroxyl group (OH). On the right is ethanol, which has two carbon atoms. It features a chain of two carbon atoms, each bonded to hydrogen atoms, and the last carbon is bonded to a hydroxyl group (OH). Both structures are labeled with their respective names, methanol and ethanol, below each formula.

[Generated by AI]

Once there are three or more carbon atoms in the parent chain, a number is used to show the position of the hydroxy (hydroxyl) functional group. Try naming the two alcohols shown in **Figure 9**. Remember to use the lowest possible number for the position of the hydroxyl functional group. Check your answer when you have named them.

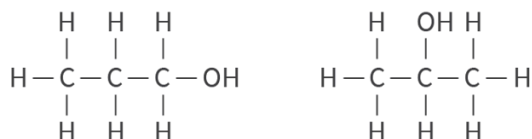


Figure 9. Try naming these two alcohols with three carbon atoms in the parent chain: remember to use the lowest possible number for the position of the OH group.

 More information for figure 9

The image displays the structural formulas of two alcohol molecules. Both molecules consist of a three-carbon atom chain (propyl chain) with hydrogen atoms attached. The first molecule on the left shows the hydroxyl group (OH) attached to the last carbon atom, illustrating a 1-propanol structure. The second molecule on the right has the hydroxyl group attached to the middle carbon atom, indicating a 2-propanol structure. Each C-H and C-O bond is shown explicitly, and the text representing the atoms is clear and legible.

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The alcohol on the left is named propan-1-ol and on the right propan-2-ol. Propan-1-ol has the hydroxy group bonded to carbon number 1 and propan-2-ol has the hydroxy group bonded to carbon number 2. Propan-3-ol does not exist because we can assign a lower number, hence it is named propan-1-ol.

So far, we have looked at naming alcohols with only one hydroxy group; there are, however, alcohols that have multiple hydroxy groups. These are known as diols (with two hydroxy groups) and triols (with three hydroxy groups). When naming diols and triols, numbers are used to signify the position of the hydroxy groups, using the lowest number possible. **Figure 10** shows an example of a diol with four carbon atoms in the parent chain that gives the parent alkane butane, which comes first in the name. There are two hydroxy groups bonded to carbon atoms numbered 1 and 3. The compound is named butane-1,3-diol. Note that the 'e' is kept on the end of the parent alkane here.

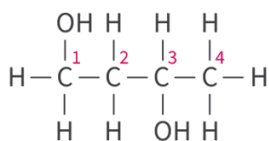


Figure 10. Butane-1,3-diol has two hydroxy groups: one bonded to carbon number 1 and one bonded to carbon number 3.

 More information for figure 10

The image is a structural diagram of butane-1,3-diol. It represents a linear chain of four carbon atoms. Each carbon atom is bonded to hydrogen atoms, except for the first and third carbon atoms, which are each bonded to a hydroxy group (OH) with labels '1' and '3' in red. The carbon atoms are connected in sequence (C-C-C-C), and the diagram shows the molecular structure visually with lines representing chemical bonds between atoms. The hydroxy groups are connected above the first and third carbon atoms, highlighting the diol nature of the compound. The numbering in red indicates the positions of these groups on the carbon chain.

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Aldehydes

Aldehydes are named by removing the 'e' from the parent alkane and replacing it with the suffix -al. The first aldehyde, with one carbon atom in the parent chain, is methanal. With two carbon atoms in the parent chain is ethanal. We then continue adding a CH₂ group to the chain to get propanal and butanal (**Figure 11**). Note that for straight-chain aldehydes a number is not required because the carbon atom of the carbonyl group is always assigned as number 1.

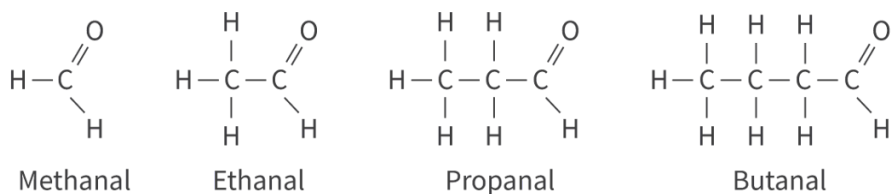


Figure 11. The full structural formulas of the first four members of the aldehyde homologous series, methanal to butanal.

 More information for figure 11

The image displays the structural formulas of the first four aldehydes. From left to right, they are: Methanal, with one carbon atom double bonded to an oxygen atom and single bonded to two hydrogen atoms (H-C(=O)H); Ethanal, with two carbon atoms, the first bonded to three hydrogen atoms and the second double bonded to an oxygen and single bonded to one hydrogen atom (CH₃-C(=O)H); Propanal, with three carbon atoms, the first and second bonded to two and one hydrogen atoms respectively, and the third carbon double bonded to an oxygen and single bonded to one hydrogen atom (CH₃-CH₂-C(=O)H); and Butanal, with four carbon atoms, where the first, second, and third carbons are each bonded to two hydrogen atoms, and the fourth carbon is double bonded to an oxygen and single bonded to one hydrogen atom (CH₃-CH₂-CH₂-C(=O)H).

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Ketones

The ketone homologous series begins with three carbon atoms in the parent chain. To name the ketones, the 'e' from the parent alkane is replaced with the suffix -one. The first member of the series is propanone and with four carbon atoms in the parent chain, we have butan-2-one (sometimes called butanone, without the '2') (**Figure 12**).

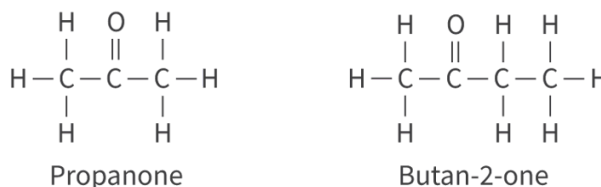


Figure 12. Propanone (left) and butan-2-one (right), the first two members of the ketone homologous series.

 More information for figure 12

The image shows the structural diagrams of two ketones, propanone on the left and butan-2-one on the right. In the propanone structure, the central carbon is double-bonded to an oxygen atom (C=O), and the other carbons are single-bonded to hydrogen atoms (H-C-H). The butan-2-one structure has a similar central carbonyl group (C=O), but it includes an additional carbon in the chain, making it four carbons long. Each diagram labels the molecule's name below it. These structures depict the basic chemical structure of ketones in their homologous series.

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Try to name the two ketones shown in **Figure 13** with five carbon atoms in the parent chain. Remember to use the lowest possible number for the position of the carbonyl group in the molecule. Check your answer when you have named them.

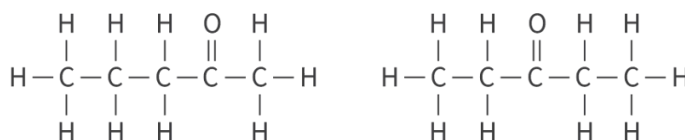


Figure 13. Try to name the two ketones with five carbon atoms in the parent chain. Remember to use the lowest possible number for the position of the carbonyl group.

 More information for figure 13

The image shows two structural formulas of ketone molecules, each with a five-carbon atom chain. The first ketone features a carbon chain aligned horizontally with the carbonyl group (C=O) bonded to the third carbon atom from the left. The chain can be expressed as: $\text{H}_3\text{C}-\text{CH}_2-\text{CO}-\text{CH}_2-\text{CH}_3$. The second ketone has a similar five-carbon

chain with the carbonyl group bonded to the second carbon atom from the left, expressed as: $\text{H}_3\text{C}-\text{CO}-\text{CH}_2-\text{CH}_2-\text{CH}_3$. Each carbon atom is bonded with hydrogen atoms to satisfy the tetravalency of carbon.

[Generated by AI]

On the left is pentan-2-one and on the right, pentan-3-one. Pentan-2-one has the carbonyl group bonded to carbon number 2 and pentan-3-one has the carbonyl group bonded to carbon number 3.

Ketones can also exist with multiple carbonyl groups, which are known as diones (with two carbonyl groups) and triones (with three carbonyl groups). **Figure 14** shows an example of a dione with five carbon atoms in the parent chain, which gives the parent alkane pentane. Numbers are used to show the position of the carbonyl groups, assigning them to the lowest possible numbers. The compound is named pentane-2,4-dione.

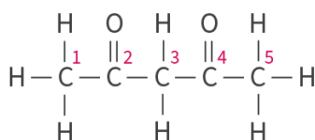


Figure 14. Pentane-2,4-dione has two carbonyl groups: one bonded to carbon number 2 and the other bonded to carbon number 4.

 More information for figure 14

The image is a structural diagram of pentane-2,4-dione, which is a five-carbon chain molecule with two carbonyl groups. The diagram displays the molecular structure with carbon atoms numbered 1 through 5. Carbon atom 1 has three hydrogen atoms attached. Carbon atoms 2 and 4 each have a double-bonded oxygen atom (carbonyl group), and single hydrogen atoms are bonded to the remaining available sites on carbons 3 and 5. The numbering indicates the positions of carbonyl groups on the carbon chain, consistent with the naming convention of the compound.

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Carboxylic acids

Carboxylic acids are named by replacing the 'e' of the parent alkane with the suffix -oic acid. The first member of the carboxylic acid series is methanoic acid followed by ethanoic acid, propanoic acid and butanoic acid (**Figure 15**). A number is not required because the carbon atom of the carboxyl functional group is always assigned as carbon number 1.

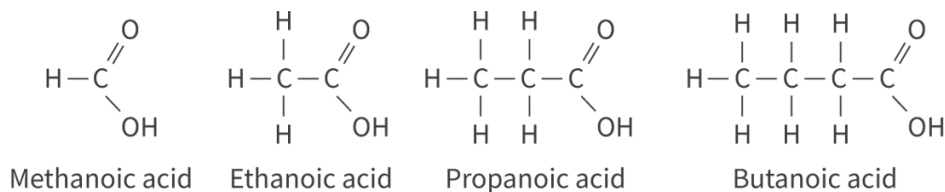


Figure 15. The first four members of the carboxylic acid homologous series, methanoic acid to butanoic acid.

More information for figure 15

The image displays the chemical structures of the first four members of the carboxylic acid homologous series: methanoic acid, ethanoic acid, propanoic acid, and butanoic acid. Each compound is represented by its structural formula.

1. **Methanoic acid:** The formula consists of a single carbon atom bound to an OH group and a double-bonded O atom, forming the simplest carboxylic group.

2. **Ethanoic acid:** The structure includes a two-carbon chain with the OH and double-bonded O attached to the first carbon.

3. **Propanoic acid:** This compound displays a three-carbon chain, with the carboxyl group at the end.

4. **Butanoic acid:** The structure shows a four-carbon chain with the carboxyl group at the terminal position.

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Carboxylic acids can also have two carboxyl groups in the molecule: these are known as dicarboxylic acids. They are named by counting the number of carbon atoms in the parent chain and adding the suffix -dioic acid. **Figure 16** has an example with three carbon atoms in the parent chain which gives the parent alkane propane. Adding the suffix -dioic acid, we get the name propanedioic acid.

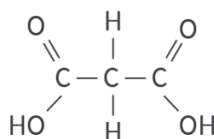


Figure 16. Propanedioic acid is a dicarboxylic acid with a carboxyl group on either end of the molecule.

More information for figure 16

The image is a structural diagram of propanedioic acid, also known as malonic acid, which is a simple dicarboxylic acid. The diagram shows three carbon atoms in a linear chain. Each of the terminal carbon atoms is double-bonded to an oxygen atom (forming a carboxyl group), and each is also bonded to an OH (hydroxyl) group. The central carbon atom is bonded to two hydrogen atoms. The diagram illustrates the symmetric nature of this molecule, with carboxyl groups positioned at each end.

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Halogenoalkanes

Halogenoalkanes are named according to the number of carbon atoms in the parent chain that gives the name of the parent alkane. A prefix is used that indicates the type of halogen atom in the compound together with a number that indicates to which carbon atom of the parent chain the halogen atom is bonded. The prefixes for the halogen atoms are shown in **Table 2**.

Table 2. Prefixes for halogens.

Halogen atom	Prefix
Fluorine	Fluoro-
Chlorine	Chloro-
Bromine	Bromo-
Iodine	Iodo-

The first example we will look at is a halogenoalkane with one carbon atom in the parent chain that gives the parent alkane –methane. The halogen atom is a chlorine atom so the prefix used is chloro-. A number is not required because there is only one carbon atom in the parent chain. The compound is named chloromethane (**Figure 17**). If the chlorine atom is replaced with a bromine atom, the compound is named bromomethane.

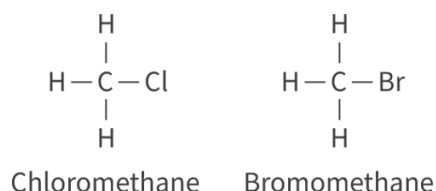


Figure 17. Full structural formulas of two halogenoalkanes with one carbon atom in the parent chain: chloromethane (left) and bromomethane (right).

 More information for figure 17

The image contains structural formulas for two halogenoalkanes: chloromethane and bromomethane. On the left, chloromethane is shown with a central carbon atom bonded to three hydrogen atoms and one chlorine atom. The formula is presented as H—C—Cl. On the right, bromomethane is depicted with a central carbon atom bonded to three hydrogen atoms and one bromine atom, shown as H—C—Br. Below each formula, the compound names 'Chloromethane' and 'Bromomethane' are written respectively, indicating the substitution of a chlorine atom in chloromethane and a bromine atom in bromomethane.

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Moving on to two carbon atoms in the parent chain and with an iodine atom in the compound, we have iodoethane (**Figure 18**). A number is not yet required because there are only two carbon atoms in the parent chain.

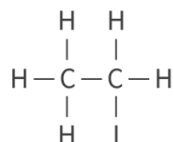


Figure 18. The full structural formula of iodoethane.

 More information for figure 18

The image is a structural diagram of iodoethane, showing its molecular structure. It consists of a two-carbon parent chain. Each carbon atom (C) is bonded to three hydrogen atoms (H) in such a way that each carbon is connected to its neighboring carbon via a single line, representing a covalent bond. The first carbon atom has two hydrogen atoms attached in a vertical manner, one at the top and one at the bottom. The second carbon atom also has a vertical alignment, with an iodine atom (I) attached at the bottom instead of a hydrogen atom. The three hydrogen atoms are distributed with one above and one bonded to each carbon, maintaining the tetrahedral geometry. The iodine atom substitutes one hydrogen atom in this structure, which characterizes it as iodoethane.

[Generated by AI]

Once we reach three carbon atoms in the parent chain, we need to use a number to indicate to which carbon atom of the parent chain the halogen atom is bonded. Take for example the compound shown in **Figure 19**: it has three carbon atoms in the parent chain and a fluorine atom bonded to the second carbon atom. Try naming it yourself before checking your answer below.

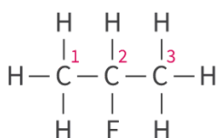


Figure 19. Name this compound: there are three carbon atoms in the parent chain and a fluorine atom bonded to carbon number 2.

 More information for figure 19

The image shows a structural diagram of a chemical compound. It features a three-carbon (propane) parent chain arranged horizontally. Each carbon atom is singly bonded to hydrogen atoms to satisfy the tetravalency of carbon. The first carbon (leftmost) is bonded to three hydrogen atoms, the second carbon is bonded to two

hydrogen atoms and a fluorine atom, and the third carbon (rightmost) is bonded to three hydrogen atoms. The carbon chain is numbered from 1 to 3 starting from the left. The fluorine atom is attached to carbon number 2.

[Generated by AI]

There are three carbon atoms in the parent chain, which gives the parent alkane –propane. There is a fluorine atom bonded to carbon number 2, which gives the prefix fluoro-. The compound is named 2-fluoropropane.

Next, we will look at halogenoalkanes that have multiple halogen atoms in the molecule. The halogen atoms can be the same or different. For example, **Figure 20** shows a halogenoalkane with three carbon atoms in the parent chain and two chlorine atoms bonded to carbon atoms one and two. The molecule is named 1,2-dichloropropane.

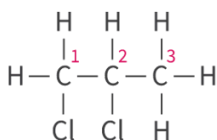


Figure 20. 1,2-dichloropropane has two chlorine atoms bonded to the carbon atoms numbered 1 and 2.

 More information for figure 20.1

The image is a structural diagram of 1,2-dichloropropane, a halogenoalkane. The molecule has a chain of three carbon atoms, numbered 1, 2, and 3 from left to right. Each carbon atom is connected to hydrogen and chlorine atoms. The first carbon atom (1) is bonded to two hydrogen atoms and one chlorine atom below it. The second carbon atom (2) is also bonded to two hydrogen atoms and one chlorine atom below it. The third carbon atom (3) is bonded to two hydrogen atoms. The diagram shows the arrangement of atoms and the bonds between them, indicating the position of the chlorine atoms on the first and second carbon atoms in the sequence.

[Generated by AI]

In the next example (**Figure 21**) we have a halogenoalkane with two different halogen atoms. These are a bromine atom and a chlorine atom. The carbon atoms are numbered so that the halogen that comes first in alphabetical order is assigned the lowest number. So, this compound is named 1-bromo-3-chloropropane.

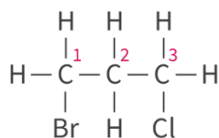


Figure 21. 1-bromo-3-chloropropane has a bromine atom bonded to carbon number 1 and chlorine atom bonded to carbon number 3.

More information for figure 21.1

The image is a chemical diagram depicting the molecular structure of 1-bromo-3-chloropropane. It consists of a linear chain of three carbon atoms (C) labeled 1, 2, and 3 from left to right. The first carbon atom (C-1) is bonded to one hydrogen atom (H) and one bromine atom (Br). The second carbon atom (C-2) is bonded to two hydrogen atoms. The third carbon atom (C-3) is bonded to one hydrogen atom and one chlorine atom (Cl). The structure is arranged so that each carbon atom forms a total of four bonds, showing the typical tetrahedral arrangement in organic molecules.

[Generated by AI]

The next example (**Figure 22**) is a halogenoalkane with five carbon atoms in the parent chain and three halogen atoms: a bromine atom, a chlorine atom and a fluorine atom. Try naming this compound for yourself before checking the answer below.

In alphabetical order, bromine is first (bonded to carbon number 2), chlorine second (bonded to carbon number 4) and fluorine last (bonded to carbon number 3). The compound is named 2-bromo-4-chloro-3-fluoropentane.

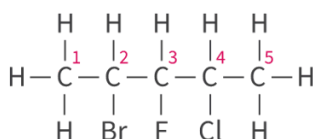


Figure 22. Try to name this halogenoalkane for yourself before checking your answer.

More information for figure 22

The image shows the structural formula of a halogenoalkane molecule. The backbone consists of a carbon chain with five carbon atoms labeled 1 through 5. Each carbon atom is bonded with hydrogen atoms to complete its valency. The first carbon is bonded to three hydrogen atoms. The second carbon is bonded to two hydrogen atoms and one bromine (Br) atom. The third carbon is bonded to two hydrogen atoms and one fluorine (F) atom. The fourth carbon is bonded to one hydrogen atom and one chlorine (Cl) atom. The fifth carbon is bonded to two hydrogen atoms. Overall, this structure represents a halogenoalkane with bromine, fluorine, and chlorine substituents on a five-carbon linear chain.

[Generated by AI]

Drag and drop the structures in **Interactive 1** to practise naming the compounds covered so far in this section. The structures are spread across two tables – use the bottom arrow to see the other screen.

Functional group structure	IUPAC name
$ \begin{array}{c} \text{H} & \text{H} & \text{H} \\ & & \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{H} \\ & & \\ \text{H} & \text{H} & \text{H} \end{array} $	1-bromo-3-chloro-2-iodobutane But-1-ene Butan-2-ol
$ \begin{array}{c} \text{H} & \text{H} & \text{H} & \text{H} \\ & & & \\ \text{H}-\text{C}=\text{C}-\text{C}-\text{C}-\text{H} \\ & & & \\ & & \text{H} & \text{H} \end{array} $	Butanedioic acid 1-chlorobutane
$ \begin{array}{c} & & \text{H} \\ & & \\ \text{H}-\text{C}\equiv\text{C}-\text{C}-\text{H} \\ & & \\ & & \text{H} \end{array} $	Ethanal
$ \begin{array}{c} \text{H} & \text{H} & \text{OH} & \text{H} \\ & & & \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C}-\text{H} \\ & & & \\ \text{H} & \text{H} & \text{H} & \text{H} \end{array} $	Ethane-1,2-diol Ethanoic acid
$ \begin{array}{c} \text{H} & \text{H} \\ & \\ \text{HO}-\text{C}-\text{C}-\text{OH} \\ & \\ \text{H} & \text{H} \end{array} $	Hexane-2,3,4-trione Propane
$ \begin{array}{c} \text{H} & & \text{O} \\ & & \\ \text{H}-\text{C}-\text{C} \\ & & \backslash \\ \text{H} & & \text{OH} \end{array} $	Propanone Propyne

☒ Check
 ➤



Interactive 1. Name the Compounds.

 More information for interactive 1

An interactive table features the functional group structures and IUPAC names of the chemical compounds. Users need to drag and drop the correct IUPAC name into the appropriate drop zones that correspond to each compound's structure.

Structures:

Carbon, oxygen, chlorine, hydrogen, and atoms and hydroxy groups are represented by their symbols.

1. Three C atoms are arranged in a straight line, connected by single bonds. The first C atom is single-bonded to H atoms at the top, left, and bottom. The second C atom is single-bonded to H atoms at the top and bottom. The third C atom is single-bonded to H atoms at the top, right, and bottom.
2. Four C atoms are arranged in a straight line, connected by single and double bonds. The first C atom is double-bonded to the second C atom, while the other C atoms are connected by single bonds. The first C atom is single-bonded to H atoms at the top and left. The second C atom is single-bonded to one H atom at the top. The third C atom is single-bonded to H atoms at the top and bottom. The fourth C atom is single-bonded to H atoms at the top, right, and bottom.
3. Three C atoms are arranged in a straight line, connected by single and triple bonds. The first C atom is triple-bonded to the second C atom, while the other C atom is connected by a single bond. The first C atom is single-bonded to one H atom at the left. The third C atom is single-bonded to H atoms at the top, right, and bottom.
4. Four C atoms are arranged in a straight line, connected by single bonds. The first C atom is single-bonded to H atoms at the top, left, and bottom. The second C atom is single-bonded to H atoms at the top and bottom. The third C atom is single-bonded to an OH group at the top and an H atom at the bottom. The fourth C atom is single-bonded to H atoms at the top, right, and bottom.
5. Two C atoms are connected to each other by a single bond. Both C atoms are each single-bonded to H atoms at the top and bottom. The first C atom is single-bonded to an OH group on the left, whereas the second C atom is single-bonded to an OH group on the right.
6. Two C atoms are connected to each other by a single bond. The first C atom is single-bonded to H atoms at the top, left, and bottom. The second C atom is double-bonded to an O atom on the top right and single-bonded to an OH group on the bottom right.
7. Four C atoms are present in the chain and all are connected by single bonds. First C atom is double-bonded to an O atom and is single-bonded to the OH group on the left and to the second C atom on the right. The second and third C atoms are connected with two H atoms using a single bond on top and bottom. The fourth C atom is double-bonded with O and single-bonded with OH group.
8. Two C atoms are present in the chain and are connected by single bonds. The first C atom is single-bonded to three hydrogen atoms on top, bottom and left. The second C atom is double-bonded to oxygen and single-bonded to hydrogen.

9. Three carbon atoms are single-bonded in a straight line. The second C is double bonded to O and single bonded to C on the left and right side. The first and third C atoms are single-bonded to three hydrogen atoms on top, bottom and one on side.
10. Six C atoms are bonded in a straight line using single bonds. First and last C atoms are single bonded with three hydrogen atoms on top, bottom and side. The second, third, fourth C atoms are double-bonded to O. The fifth C atom is single-bonded to two H on top and bottom.
11. Four C atoms are single-bonded in a straight line. The first C atom is single-bonded to H on top and left and Cl on bottom. The second and third C are single-bonded with two H atoms on top and bottom. The fourth C is single-bonded to three H atoms on top, bottom and right.
12. Four C atoms are single-bonded in a straight line. The first C atom is single-bonded to H on top and left and Br on bottom. The second C is single-bonded to H on top and I on bottom. The third C is single-bonded to H on top and Cl on bottom. The fourth C is single-bonded to three H atoms on top, bottom and right.

Drag-and-drop options are as follows:

1-bromo-3-chloro-2-iodobutane
But-1-ene
Butan-2-ol
Butanedioic acid
1-chlorobutane
Ethanal
Ethane-1,2-diol
Ethanoic acid
Hexane-2,3,4-trione
Propane
Propanone
Propyne

This interactivity helps users understand how to apply IUPAC rules to correctly name organic compounds.

Read below for the solution:

1. Propane
2. But-1-ene
3. Propyne
4. Butan-2-ol
5. Ethane-1,2-diol
6. Ethanoic acid
7. Butanedioic acid
8. Ethanal
9. Propanone

10. Hexane-2,3,4-trione
11. 1-chlorobutane
12. 1-bromo-3-chloro-2-iodobutane

Study skills

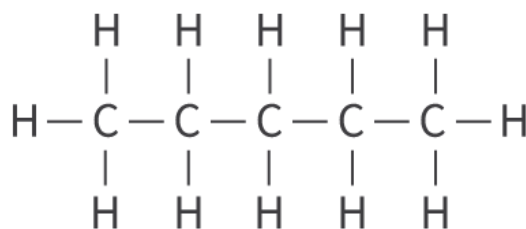
In this section, you have learned how to name organic compounds using IUPAC nomenclature, but be aware that you should also know how to sketch an organic compound from its IUPAC name.

In this activity, you will practise naming a variety of organic compounds from different types of structure.

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 30 minutes
- **Activity type:** Individual activity

In this activity, you will practise naming organic compounds using IUPAC nomenclature rules.

 Turn

Card 1 of 11



Interactive 2. Naming Organic Compounds from the Full Structural Formula.

 More information for interactive 2

An interactive carousel selector features eleven pluck cards displaying the chemical structures of organic compounds, requiring users to identify the name of each structure.

Users can switch between the cards using arrows present in the bottom right and bottom left. They can check the answer by selecting a turn button at the bottom of each card.

In the structures, carbon, oxygen, iodine, hydrogen, and fluorine atoms and hydroxy groups are represented by their symbols.

Card 1:

Five C atoms are arranged in a straight line, connected by single bonds. Each C atom is single-bonded to H atoms at the top and bottom. The first and second C atoms are each single-bonded to one H atom.

Card 2:

Five C atoms are arranged in a straight line, connected by single and double bonds. The first C atom is double-bonded to the second C atom, while the remaining C atoms are connected by single bonds. The first C atom is single-bonded to H atoms at the top and left. The second C atom is single-bonded to one H atom at the top. The third, fourth, and fifth C atoms are each single-bonded to H atoms at the top and bottom. The fifth C atom is also single-bonded to one H atom on the right.

Card 3:

Five C atoms are arranged in a straight line, connected by single and triple bonds. The second C atom is triple-bonded to the third C atom, while the remaining C atoms are connected by single bonds. The first C atom is single-bonded to H atoms at the top, bottom, and left. The fourth and fifth C atoms

are each single-bonded to H atoms at the top and bottom. The fifth C atom is also single-bonded to one H atom on the right.

Card 4:

Four C atoms are arranged in a straight line, connected by single bonds. Each C atom is single-bonded to H atoms at the top and bottom. The first C atom is also single-bonded to one H atom on the left. The fourth carbon atom is single-bonded to an OH group on the right.

Card 5:

Five C atoms are arranged in a straight line, connected by single bonds. The first four C atoms are each single-bonded to H atoms at the top and bottom. The first C atom is also single-bonded to one H atom on the left. The fifth carbon atom is double-bonded to an O atom and single-bonded to an H atom.

Card 6:

Six C atoms are arranged in a straight line, connected by single bonds. The first, second, fourth, fifth, and sixth C atoms are each single-bonded to H atoms at the top and bottom. The first C atom is also single-bonded to one H atom on the left. The sixth C atom is single-bonded to one H atom on the right. The third C atom is double-bonded to an O atom on the top.

Card 7:

Five C atoms are arranged in a straight line, connected by single bonds. The first four C atoms are each single-bonded to H atoms at the top and bottom. The first C atom is also single-bonded to one H atom on the left. The fifth C atom is double-bonded to an O atom and single-bonded to an OH group.

Card 8:

Five C atoms are arranged in a straight line, connected by single bonds. The second, third, and fourth C atoms are each single-bonded to H atoms at the top and bottom. The first and fifth C atoms are each double-bonded to an O atom and single-bonded to an OH group.

Card 9:

Three C atoms are arranged in a straight line, connected by single bonds. The first and third C atoms are each single-bonded to H atoms at the top and bottom. The first C atom is also single-bonded to one H atom on the left, and the third C atom is also single-bonded to one H atom on the right. The second carbon is single-bonded to one H atom at the top and one I atom at the bottom.

Card 10:

Four C atoms are arranged in a straight line, connected by single bonds. The first C atom is single-bonded to H atoms at the top and left and an F atom on the bottom. The second C atom is single-bonded to H atoms at the top and bottom. The third C atom is single-bonded to an H atom at the top and an F atom at the bottom. The fourth C atom is single-bonded to H atoms at the top, bottom, and right.

Card 11:

Two C atoms are arranged in a straight line, connected by a single bond. Each C atom is single-bonded to H atoms at the top and bottom. The first C atom is single-bonded to an OH group on the left, and the second C atom is single-bonded to an OH group on the right.

This interactivity helps users identify organic compounds, improve structural visualisation skills, and enhance understanding of chemical nomenclature.

Read below for the solution:

Card 1: Pentane

Card 2: Pent-1-ene

Card 3: Pent-2-yne

Card 4: Butan-1-ol

Card 5: Pentanal

Card 6: Hexan-3-one

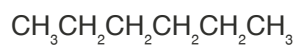
Card 7: Pentanoic acid

Card 8: Pentanedioc acid

Card 9: 2-iodopropane

Card 10: 1,3-difluorobutane

Card 11: Ethane-1,2-diol



↺ Turn

Card 1 of 8



Interactive 3. Naming organic compounds from the condensed structural formula.

 Turn

Card 1 of 8



Interactive 4. Naming Organic Compounds from the Skeletal Formula.

 More information for interactive 4

An interactive carousel selector features eight pluck cards displaying the skeletal structures of organic compounds, requiring users to identify the name of each structure.

Users can switch between the cards using arrows present in the bottom right and bottom left. They can check the answer by selecting a turn button at the bottom of each card.

In the structures, the zigzag line represents the carbon backbone, and hydrogen atoms are implied but not shown. Each endpoint and vertex of the zigzag represents a carbon atom.

Card 1:

A zigzag line represents a four-carbon chain.

Card 2:

A zigzag line represents a six-carbon chain. A double bond is present between the first and second carbon atoms from the right.

Card 3:

A zigzag line represents a six-carbon chain. A triple bond is present between the first and second carbon atoms from the right. The triple bond is shown as a linear structure.

Card 4:

A zigzag line represents a six-carbon chain. An OH group is single-bonded to the second carbon atom from the left.

Card 5:

A zigzag line represents a four-carbon chain. An O atom is double-bonded to the second carbon atom from the left.

Card 6:

A zigzag line represents a four-carbon chain. An O atom is double-bonded to the first carbon atom from the left.

Card 7:

A zigzag line represents a four-carbon chain. An O atom is double-bonded to the first carbon atom from the left. Also, an OH group is single-bonded to the same carbon atom.

Card 8:

A zigzag line represents a five-carbon chain. A Br atom is single-bonded to the second carbon atom from the left.

This interactivity helps users recognise and interpret skeletal structures of organic compounds.

Read below for the solution:

Card 1: Butane

Card 2: Hex-1-ene

Card 3: Hex-1-yne

Card 4: Hexan-2-ol

Card 5: Butanone or butan-2-one

Card 6: Butanal

Card 7: Butanoic acid

Card 8: 2-bromopentane

Nomenclature of branched-chain organic compounds

Learning outcomes

At the end of this section you should be able to apply IUPAC nomenclature to saturated or mono-unsaturated compounds that have up to six carbon atoms in the parent chain and contain one type of the following functional groups: halogeno, hydroxy (hydroxyl), carbonyl, carboxyl.

In this section, you will learn how to name branched-chain organic compounds. These are compounds with branches or alkyl groups branching off from the parent chain. To name branched-chain compounds, the following rules are applied:

1. Identify the longest continuous carbon chain in the molecule (the parent chain). This gives the name of the parent alkane.
2. Identify and name the side-chains or substituent groups; these give the prefix of the name, as shown in **Table 1**. If there is more than one side-chain of the same type, use the prefixes di-, tri-, tetra- and so on. If a molecule contains several different side-chains, list them in alphabetical order.
3. Identify the position of the side-chain(s) using a number that refers to the number of the carbon atom in the chain. The carbon atoms in the parent chain are numbered starting from the end, which will give the substituent groups the lowest number.

Table 1. Names for branched-chain organic compounds.

Number of carbon atoms in side-chain/branch	Name of side-chain/branch	Condensed structure
1	methyl	—CH ₃
2	ethyl	—CH ₂ CH ₃
3	propyl	—CH ₂ CH ₂ CH ₃

Number of carbon atoms in side-chain/branch	Name of side-chain/branch	Condensed structure
4	butyl	$-\text{CH}_2\text{CH}_2\text{CH}_2\text{C}$

Study skills

When naming organic compounds, commas are used to separate numbers and dashes are used to separate numbers and letters. For example, the organic compound 2,2-dimethylbutane has a comma between the two '2's and a dash between the '2' and the 'dimethylbutane'.

First, we will look at naming branched-chain alkanes. **Figure 1** shows two branched-chain alkanes with the same molecular formula, C_5H_{12} . As the next section also shows, these are known as structural isomers.

- The compound on the left has four carbon atoms in the parent chain, which gives the parent alkane butane. There is a methyl group ($-\text{CH}_3$) bonded to the second carbon atom in the parent chain. The compound is named 2-methylbutane and has the condensed structural formula $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_3$ (note the branches are included inside brackets).
- The compound on the right has three carbon atoms in the parent chain with two methyl groups that are both bonded to the second carbon atom. The compound is named 2,2-dimethylpropane (condensed structural formula $\text{CH}_3\text{C}(\text{CH}_3)_2\text{CH}_3$).

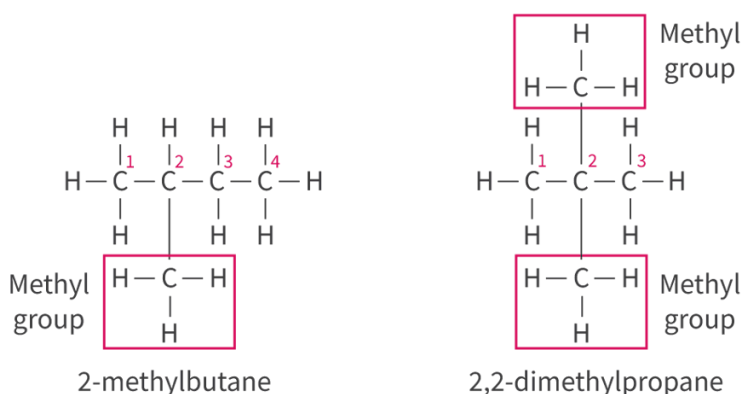


Figure 1. The branched-chain alkanes 2-methylbutane (left) and 2,2-dimethylpropane (right).

More information for figure 1

The image shows the chemical structures of two branched-chain alkanes: 2-methylbutane on the left and 2,2-dimethylpropane on the right.

2-Methylbutane Structure: - The main carbon chain consists of four carbon atoms in a straight line. These are numbered from left to right as 1, 2, 3, and 4. - Carbon 2 is connected to a CH₃ group (methyl group), indicated in the image. - All other hydrogens are arranged to satisfy the tetravalence of carbon.

2,2-Dimethylpropane Structure: - The central carbon is surrounded by three carbons, forming a tetrahedral structure. - The central carbon is bonded to two CH₃ groups (methyl groups) on the top and bottom. - Carbon atoms are numbered from the central carbon outward as 1, 2, and 3 for each chain extending to the left and right of the central carbon. - Hydrogens are arranged around the carbon atoms to maintain chemical integrity.

[Generated by AI]

Identifying the parent chain can be a challenge in some branched compounds, as can be seen in the next example. Try to find the parent chain in the compound shown in **Figure 2**. Once you have done so, try to identify the branches. Check your answer once you are ready.

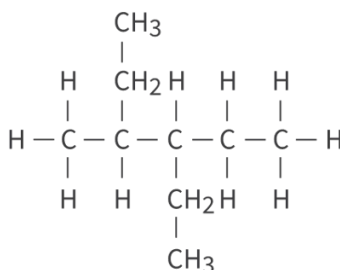
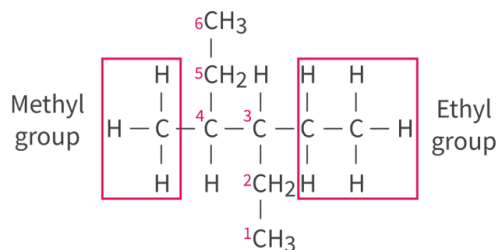


Figure 2. Try to identify the parent chain and the branches in this branched alkane.

 More information for figure 2

This is a structural diagram of a branched alkane. It consists of a central horizontal chain of five carbon atoms, each bonded with hydrogen atoms. Starting from the left, the first carbon (C) is single-bonded to three hydrogen (H) atoms. The second carbon is bonded to two hydrogens. The third carbon forms part of a branch: it is bonded to two hydrogens and a CH₃ (methyl group), which branches above. The fourth carbon is similar, bonded to two hydrogens and has another CH₃ group attached below. The final carbon on the right is bonded to three hydrogens. This forms a backbone structure with methyl groups branching off the third and fourth carbon atoms, illustrating a typical branched alkane structure.

[Generated by AI]



The parent chain is six carbon atoms with a methyl group ($-\text{CH}_3$) and an ethyl group ($-\text{C}_2\text{H}_5$) as branches. Notice how the parent chain is numbered giving the ethyl group the lowest number. This is because the branches are numbered according to the alphabetical order: ethyl comes before methyl so it is assigned the lowest number and comes first in the name. This compound is named 3-ethyl-4-methylhexane.

Worked example 1

Deduce the IUPAC name of the following branched-chain alkane.

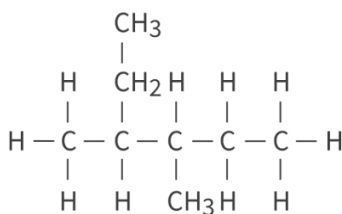


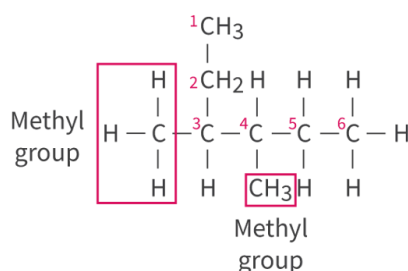
Figure 3. A branched-chain alkane.

More information for figure 3

The image depicts the chemical structure of a branched-chain alkane. The structure features a main carbon chain consisting of five carbon atoms, with hydrogen atoms attached along the chain. A methyl group (CH_3) is attached to the second carbon from the left. The main chain can be described from left to right as follows: starting with a carbon bonded to three hydrogens, then a carbon bonded to two hydrogens and one carbon of a methyl group, followed by three more carbons each bonded to two hydrogens. The second and last carbon atoms are bonded to additional hydrogens to satisfy the tetravalency of carbon. The entire structure illustrates a typical alkane with a branching methyl group, which classifies it as a branched-chain alkane.

[Generated by AI]

The parent chain is six carbon atoms long which gives the stem hexane. There are two methyl branches on carbons 3 and 4. Hence the compound is named 3,4-dimethylhexane.



Worked example 2

Deduce the IUPAC name of the following branched-chain alkane.

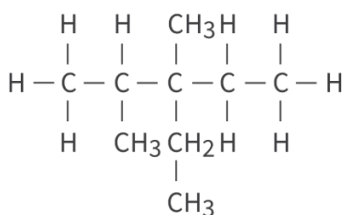


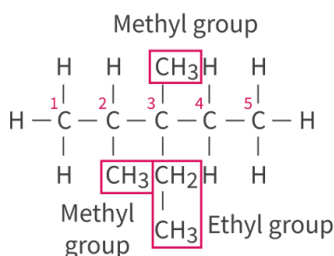
Figure 4. A branched-chain alkane.

More information for figure 4

The image is a structural formula diagram of a branched-chain alkane. The main carbon chain consists of five carbon atoms connected in a horizontal line. Each carbon atom is bonded to enough hydrogen atoms to fill its four bonds unless bonded to another carbon. The second carbon from the left is bonded to a CH₃ group below the chain, forming a branch. The third carbon is bonded to a CH₂ group below, which is further bonded to a CH₃ group, forming a longer branch. This visual structure helps in understanding the arrangement of atoms in the molecule.

[Generated by AI]

The parent chain is five carbon atoms long which gives the stem pentane. There are two methyl branches on carbons 2 and 3. There is an ethyl group on carbon number 3. The compound is named 3-ethyl-2,3-dimethylpentane.



Branched-chain alkenes and alkynes

Alkenes and alkynes can also be branched. They are named in a similar way to branched alkanes that use a number to show the position of the branch(es).

Figure 5 shows two branched alkenes with four carbons in the parent chain, which gives the parent alkenes but-1-ene and but-2-ene, respectively. Both compounds have a methyl group bonded to the second carbon atom. The names of the compounds are 2-methylbut-1-ene and 2-methylbut-2-ene.

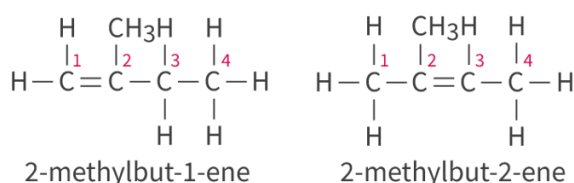


Figure 5. 2-methylbut-1-ene and 2-methylbut-2-ene are branched alkenes that differ in the position of the C=C bond.

More information for figure 5.2

The image shows the structural formulas of two branched alkenes, 2-methylbut-1-ene and 2-methylbut-2-ene. Both structures depict a parent chain of four carbon atoms. In 2-methylbut-1-ene, the double bond is between the first and second carbon atoms. There is a methyl group (CH_3) connected to the second carbon. For 2-methylbut-2-ene, the double bond is between the second and third carbon atoms, with the methyl group attached to the second carbon, similarly. Each carbon atom is bonded with hydrogen to satisfy the tetravalency of carbon. The alkene structures demonstrate the different positions of the double bond within the carbon chain, affecting the naming of these compounds.

[Generated by AI]

Next, we have two branched alkynes with five carbon atoms in the parent chain, which gives the parent alkynes pent-1-yne and pent-2-yne, respectively (**Figure 6**). Each compound has a methyl group bonded to carbon number 3 or carbon number 4. The compounds are named 3-methylpent-1-yne and 4-methylpent-2-yne.

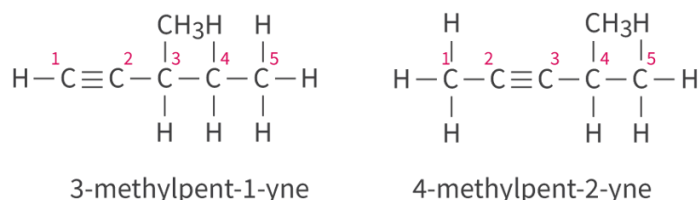


Figure 6. 3-methylpent-1-yne and 4-methylpent-2-yne differ in the positions of both the C=C bond and the CH₃ group.

 More information for figure 6.3

The image shows two chemical structures for the compounds 3-methylpent-1-yne and 4-methylpent-2-yne. Both diagrams have carbon atoms labeled from 1 to 5.

For 3-methylpent-1-yne: - The structure starts with a hydrogen bonded to a carbon triple-bonded to another carbon. - The second carbon is bonded to a carbon in the third position, which is also bonded to a methyl group (CH₃) above it and a hydrogen below it. - The chain continues with two more carbon atoms each bonded to two hydrogens.

For 4-methylpent-2-yne: - The structure begins with a hydrogen bonded to a carbon, followed by a single bond to another carbon. - The second carbon is triple-bonded to a carbon in position three. - The third carbon is single-bonded to another carbon in position four. - A methyl group (CH₃) is connected to the fourth carbon above it. - The chain completes with a single-bonded carbon and associated hydrogens in both structures.

[Generated by AI]

Next, you will learn how to name branched compounds that contain a functional group. When naming these compounds, the parent chain is numbered so that the functional group has the lowest number. The first example is shown in **Figure 7**. Here, we have a branched carboxylic acid that has three carbon atoms in the parent chain and a methyl group bonded to carbon number 2. Note that we start numbering the parent chain from the carbon atom of the carboxyl group, which is assigned the number one.

- There are three carbon atoms in the parent chain so the parent alkane is propane.
- The functional group is a carboxyl group so the suffix is –anoic acid.
- The prefix is the position and name of the branch, a methyl group, which is bonded to carbon number 2.

The compound is named 2-methylpropanoic acid.

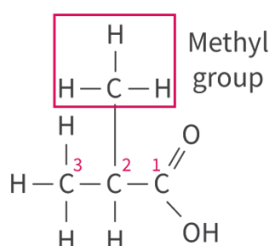


Figure 7. The branched carboxylic acid 2-methylpropanoic acid.

 More information for figure 7

This image is a diagram showing the chemical structure of 2-methylpropanoic acid. The structure includes a main chain of three carbon atoms, numbered as 1, 2, and 3. Carbon 1 is bonded to a carboxylic acid group, consisting of a double-bonded oxygen and a hydroxyl group (OH). Carbon 2 is connected to three hydrogen atoms. Carbon 3 has two hydrogen atoms and is bonded to a methyl group, labeled in a rectangular box. This methyl group is a branch, consisting of one additional carbon connected to three hydrogen atoms.

[Generated by AI]

The next example is a branched ketone with five carbon atoms in the parent chain (**Figure 8**). When naming this compound, we start numbering the carbon atoms that give the carbonyl group the lowest number, which is the number 2. There is an ethyl group bonded to carbon number 3. The compound is named 3-ethylpentan-2-one. Note that when identifying the parent chain, it must contain the functional group: in this case, this is the carbonyl group.

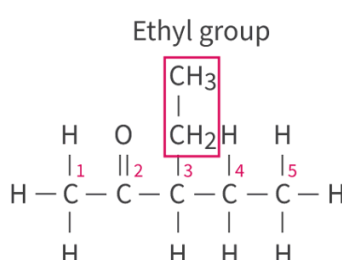


Figure 8. 3-ethylpentan-2-one is a branched ketone with five carbon atoms in the parent chain.

 More information for figure 8.3

The image shows the structural formula of 3-ethylpentan-2-one, a branched ketone. It consists of a five-carbon chain with the carbon atoms numbered from 1 to 5. The carbonyl group (C=O) is located on the second carbon (C2) of the chain. An ethyl group (CH₃-CH₂-) is attached to the third carbon (C3), which is highlighted in the diagram. Each carbon atom in the chain is bonded to hydrogen atoms to fulfill the tetravalency. The ethyl group is indicated with a box and labeled. The hydrogens bonded to each carbon and their positions show the overall structure of the molecule.

[Generated by AI]

Figure 9 shows a branched alcohol with three carbon atoms in the parent chain. The hydroxyl group is bonded to carbon number 1 and there is a methyl group bonded to carbon number 2. The compound is named 2-methylpropan-1-ol.

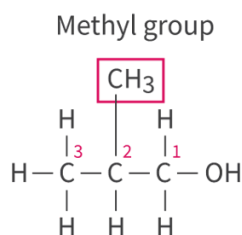


Figure 9. 2-methylpropan-1-ol is a branched alcohol with three carbon atoms in the parent chain.

 More information for figure 9.2

The image depicts the chemical structure of 2-methylpropan-1-ol. It shows a branched alcohol with a chain of three carbon atoms. The hydroxyl group (OH) is bonded to the first carbon atom, labeled as "1." Next, there is a second carbon atom labeled "2" bonded to a methyl group (CH₃) at the top, which is highlighted and labeled "Methyl group." The third carbon atom, labeled "3," completes the carbon chain. Each carbon atom in this structure is also bonded to hydrogen atoms.

[Generated by AI]

Worked example 3

Deduce the IUPAC name of the branched compound shown.

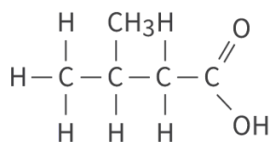


Figure 10. A branched compound.

 More information for figure 10

The image shows the structural formula of a branched organic compound. The skeleton consists of four carbon atoms in a row, with single bonds between them. The first carbon atom is bonded to three hydrogen atoms. The second carbon atom is bonded to two hydrogen atoms and a CH₃ group (methyl group) branching above. The third carbon atom is bonded to two hydrogen atoms. The fourth carbon atom forms a double bond with an oxygen atom and a single bond with an OH group (hydroxyl group), indicating a carboxylic acid functional group.

[Generated by AI]

There are four carbon atoms in the parent chain, which gives the parent alkane butane. The functional group is a carboxyl group, which gives the suffix -oic acid. There is a methyl group bonded to carbon number 3. The compound is named 3-methylbutanoic acid.

Worked example 4

Deduce the IUPAC name of the branched compound shown.

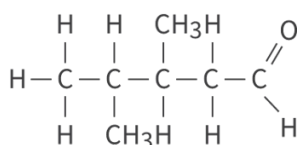


Figure 11. A branched compound.

 More information for figure 11

The image depicts the chemical structure of a branched organic compound. The main chain consists of five carbon atoms in a row, each bonded to hydrogen atoms. The compound starts with a CH₃ group, the first and second carbons each bond to two H atoms. The third carbon is bonded to a CH₃ group, forming a branch. The fourth carbon atom bonds with a carbonyl group (C=O), and a hydrogen atom completes the structure. The overall formula implies a ketone group due to the double-bonded oxygen.

[Generated by AI]

There are five carbon atoms in the parent chain, which gives the parent alkane pentane. The compound is an aldehyde, which gives the suffix -al. There are two methyl groups: one bonded to carbon number 3 and one to carbon number 4. The compound is named 3,4-dimethylpentanal.

Naming mono-unsaturated compounds

To finish this section, we will look at how to name mono-unsaturated compounds that contain one functional group. A mono-unsaturated compound is a compound that contains one C=C bond. When naming such compounds, the numbering system usually gives priority to the functional group over the C=C bond, meaning that the lowest number is assigned to the position of the functional group. There are some functional groups, however, that have lower priority than the C=C bond. **Table 2** shows the priorities of the classes and functional groups covered in the DP chemistry course from highest to lowest. Note that, with the exception of the halogenoalkanes, most of the classes or functional groups have priority over the C=C bond.

Table 2. Priorities of the classes or functional groups of organic compounds.

Class / functional group	Priority (highest to lowest)
Carboxyl	1
Aldehyde	2
Ketone	3
Hydroxyl	4
Alkene	5
Halogenoalkane	6

First you will learn how to name mono-unsaturated alcohols such as the one shown in **Figure 12**. This alcohol has two carbon atoms in the parent chain, which now gives the parent alkene (ethene) because of the presence of the C=C bond. The 'e' of the parent alkene is dropped and the suffix -ol is added. This compound is named ethenol.

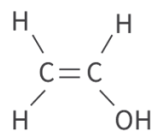


Figure 12. The full structural formula of ethenol, a mono-unsaturated alcohol with two carbon atoms in the parent chain.

 More information for figure 12

The image shows the structural formula of ethenol, a mono-unsaturated alcohol. The formula includes two carbon atoms connected by a double bond (C=C) in the center. One carbon atom is bonded to two hydrogen atoms (H), while the second carbon is bonded to one hydrogen atom and a hydroxyl group (OH). This structure represents the chemical ethenol, showcasing its unsaturated nature due to the presence of the C=C double bond. The arrangement demonstrates the typical structure for naming alcohols in organic chemistry based on the number of carbon atoms and the functional groups.

[Generated by AI]

The next example in **Figure 13** has three carbon atoms in the parent chain that gives the parent alkene propene. The C=C bond is between carbon atoms 2 and 3, so it changes to prop-2-ene. The 'e' from the parent alkene is dropped and the suffix -ol is added. The hydroxyl group is assigned the lower number as it has priority over the C=C bond so is given the number 1. This gives the name prop-2-en-1-ol.

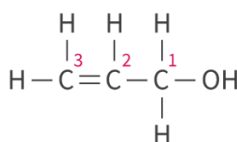


Figure 13. Prop-2-en-1-ol has three carbon atoms in the parent chain and a C=C between carbon atoms 2 and 3.

 More information for figure 13

The image shows a structural formula for prop-2-en-1-ol. It consists of a chain of three carbon atoms. Beginning from the left, Carbon 3 is involved in a double bond (C=C) with Carbon 2. Carbon 1, on the right, is bonded with a hydroxyl group (OH) and is indicated with a priority due to the functional group. The hydrogen atoms are attached to the carbon atoms: two hydrogens are attached to Carbon 3, one hydrogen is attached to Carbon 2, and one hydrogen and a hydroxyl group are attached to Carbon 1. The sequence and structure validate the IUPAC name, indicating a double bond between carbons 2 and 3, and an alcohol group on carbon 1.

[Generated by AI]

With four carbon atoms in the parent chain, there are several structural isomers that exist. One isomer is the compound shown in **Figure 14**. The hydroxyl functional group is bonded to carbon number 1 and the C=C bond is located between carbon numbers 3 and 4. This compound is named but-3-en-1-ol.

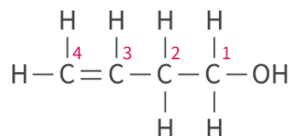


Figure 14. The full structural formula of but-3-en-1-ol. Note the OH group has priority when numbering the carbon atoms.

 More information for figure 14

This image depicts the structural formula of but-3-en-1-ol, a chemical compound. It includes a chain of four carbon atoms labeled from 1 to 4. The first carbon is bonded to a hydroxyl group (OH), indicating that it is a primary alcohol. The double bond between carbons 3 and 4 indicates the presence of an alkene group in the molecule. Additionally, each carbon atom is bonded to hydrogen atoms to satisfy the valency of carbon, with the structure suggesting single bonds for remaining valencies, except where designated by the double bond. The arrangement and bonding showcase the functional groups and molecular arrangement that define the compound's properties.

[Generated by AI]

The hydroxyl group can also be bonded to carbon number 2 as shown in **Figure 15**. Try naming this molecule for yourself before checking the answer below.

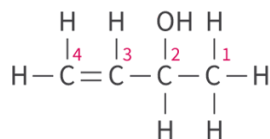


Figure 15. Try naming this compound for yourself before checking the answer below.

 More information for figure 15

The image shows a chemical structure diagram of a hydrocarbon chain with four carbon atoms numbered 1 to 4. The carbon atoms are connected in a chain. Starting from the left, carbon 1 is bonded to three hydrogen atoms. Carbon 2 is bonded to one hydrogen atom and one hydroxyl group (OH). Carbon 3 is bonded to two hydrogen atoms, and carbon 4 is bonded to three hydrogen atoms. The configuration indicates the presence of a hydroxyl group bonded to the second carbon atom. The diagram uses standard chemical notation with lines representing bonds and letters representing elements.

[Generated by AI]

This compound is named but-3-en-2-ol. The OH group is bonded to carbon number 2 and the C=C bond is between carbon numbers 3 and 4.

The last isomer of C₄H₈O is shown in **Figure 16**. The hydroxyl group is bonded to carbon number 1 and the C=C bond is between carbon number 2 and carbon number 3. Try naming this compound before checking your answer below.

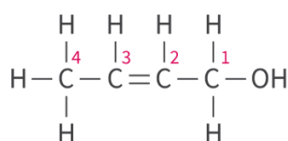


Figure 16. Try naming this compound for yourself before checking the answer below.

 More information for figure 16

The image is a structural formula diagram of a C₄H₈O compound. It consists of four carbon atoms arranged in a linear sequence. The carbon atoms are labeled from 1 to 4, starting from the right side of the image. Carbon 1 is attached to a hydroxyl group (OH) and is also connected to a carbon-carbon double bond that exists between carbon 2 and carbon 3. Additionally, each carbon atom is bonded to sufficient hydrogen atoms to satisfy the valency of carbon. The image outlines the structure necessary to represent the isomer of the compound, highlighting hydroxyl group placement and double bond positioning.

[Generated by AI]

The compound is named but-2-en-1-ol.

Next, we will look at naming mono-unsaturated carboxylic acids. To name mono-unsaturated carboxylic acids, the 'e' from the corresponding alkene is dropped and the suffix -oic acid is added. The first mono-unsaturated carboxylic acid is propenoic acid, which has three carbon atoms in the parent chain (**Figure 17**). A number is not required for propenoic acid because the carbon atom of the carboxyl group is always assigned as number 1 and the C=C bond can only be between carbon atoms 2 and 3.

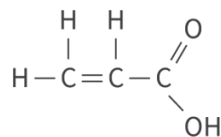


Figure 17. The full structural formula of propenoic acid. As with other examples you have seen, the carboxyl group takes priority over the C=C bond.

 More information for figure 17

The image displays the full structural formula of propenoic acid. It includes a chain of three carbon atoms. The first carbon is attached to two hydrogen atoms. The second carbon is connected by a double bond to the first carbon and is also attached to two hydrogen atoms. The third carbon is double-bonded to an oxygen atom and single-bonded to a hydroxyl group (an OH group), forming a carboxyl group. This setup highlights the structure of propenoic acid, emphasizing the priority of the carboxyl group over the carbon-carbon double bond.

[Generated by AI]

With four carbon atoms in the parent chain, there are two possible isomers. Try naming these two isomers in **Figure 18** using the rules you have learned so far. Remember to use a number to show the position of the C=C bond and to start numbering from the carbon atom of the carboxyl group. Check your answers below when you are ready.

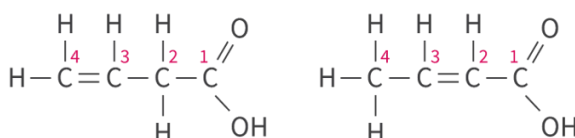


Figure 18. Try naming these two isomers for yourself before checking your answers below.

 More information for figure 18

The image shows two structural formulas of isomers, each with four carbon atoms and a carboxyl group. In each structure, carbon atoms are numbered from 1 to 4.

In the first structure (on the left), labeled as 1: The carbon chain begins with three hydrogen atoms (H) attached to the first carbon, and proceeds with a single bond between the first and second carbons. The second carbon has a double bond with the third carbon. Two hydrogen atoms are attached to the third carbon, and one hydrogen atom and a carboxyl group (OH) are attached to the fourth carbon. A double-bonded oxygen atom (O) is also attached to the same carbon.

In the second structure (on the right), labeled as 2: The carbon chain begins with two hydrogen atoms attached to the first carbon and a double bond between the first and second carbons. The third carbon atom has one hydrogen atom, and has a single bond

with the fourth carbon, which carries a double-bonded oxygen atom and a hydroxyl group (OH).

[Generated by AI]

The isomer on the left is named but-3-enoic acid and on the right is but-2-enoic acid. The numbers three and two show the position of the C=C bond in the molecule.

Moving on to mono-unsaturated aldehydes, with the first example being propenal (**Figure 19**). Notice that the 'e' is dropped from the parent alkene and the suffix -al is added. A number is not required for propenal because the carbon atom of the carbonyl group is always assigned as number 1.

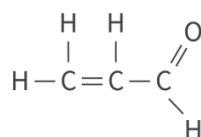


Figure 19. The full structural formula of propenal.

With four carbon atoms in the parent chain, there are two possible isomers shown in **Figure 20**. Try naming these two isomers using the rules you have learned so far. Remember that the carbon atom of the carbonyl group is assigned as number one. Check your answers below when you are ready.

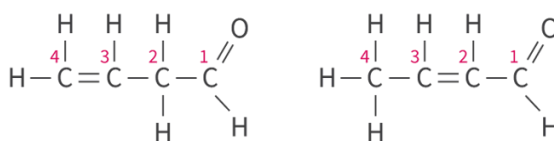


Figure 20. Try naming these isomers before checking your answer below.

 More information for figure 20

The image depicts two structural isomers, each with a four-carbon backbone. On the left isomer, the carbon atoms are connected in a linear single-double-single-single bond sequence. The carbonyl group (C=O) is present at the end of the chain. The hydrogen atoms are attached to the remaining valencies of the carbon atoms. The sequence is C1 with C=O and a single H, C2 and C3 with single H and a double bond between them, and C4 with two H.

The right isomer features a different arrangement with the carbonyl group (C=O) attached to C1, followed by a single bond to C2, a double bond between C2 and C3, and a single bond between C3 and C4. Hydrogen atoms fill the remaining valencies for

C2, C3, and C4. Carbon 2 has two hydrogens, and C3 and C4 each have one hydrogen, with the remaining valency taken up by the double bond between C2 and C3.

[Generated by AI]

The isomer on the left is but-3-enal and on the right we have but-2-enal.

Next, we have the mono-unsaturated ketones. To name mono-unsaturated ketones, the 'e' is dropped from the parent alkene and the suffix -one is added. The carbonyl group has priority over the C=C bond so we assign it the lowest number. With four carbon atoms in the parent chain, we have but-3-en-2-one (Figure 21).

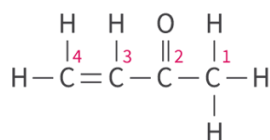


Figure 21. But-3-en-2-one has a carbonyl group bonded to carbon number 2 and a C=C bond between carbon atoms 3 and 4.

 More information for figure 21

The image illustrates the chemical structure of but-3-en-2-one. The structure consists of four carbon atoms in a linear arrangement. Carbon atoms 1 through 4 are labeled in red with their respective numbers. The first carbon atom (carbon 1) is bonded to two hydrogen atoms and is connected to the second carbon atom (carbon 2) via a single bond. Carbon 2 is connected via a double bond to an oxygen atom (forming a carbonyl group) and is connected to carbon 3 via another single bond. Carbon 3 forms a double bond with carbon 4. Carbon 3 is also bonded to one hydrogen atom, and carbon 4 is connected to two hydrogen atoms. The connectivity illustrates the molecular structure of the compound, highlighting the presence of a carbon-carbon double bond and a carbonyl group.

[Generated by AI]

With five carbon atoms in the parent chain, there are three possible isomers. The structure of one isomer is shown in Figure 22.

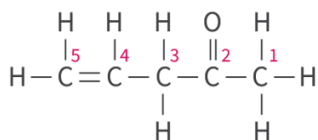


Figure 22. Pent-4-en-2-one has a carbonyl group bonded to carbon number 2 and a C=C bond between carbon atoms 4 and 5.

 More information for figure 22

The image shows the structural formula of Pent-4-en-2-one. It consists of a carbon chain with five carbon atoms:

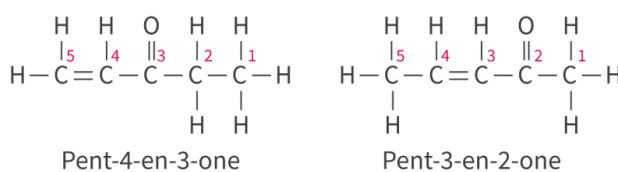
1. The first carbon (on the right) is bonded to three hydrogen atoms (C-H₃).
2. The second carbon is a carbonyl carbon featuring a C=O bond; it also is connected to a methyl group (C-CH₃).
3. The third carbon is single-bonded to two hydrogen atoms (C-H₂) and adjacent to the carbonyl group.
4. The fourth carbon is double-bonded to the fifth carbon and bonded to a single hydrogen atom (H-C=C), indicating a C=C bond.
5. The fifth carbon is bonded to two hydrogen atoms (H-C-H), completing the double bond with the fourth carbon.

The carbon atoms are numbered from right to left: 1 to 5, and important features like the double bond and the carbonyl group are highlighted.

Below the formula, numbers in red label the carbon atoms: 1 on the rightmost, followed by 2, which is the carbonyl carbon, and so on till the fifth carbon on the leftmost position. There are no colors or other specific visual details beyond what is described that are necessary for understanding.

[Generated by AI]

Try sketching out the full structural formulas of the other two isomers and naming them. Check your answers once you are ready.



The two isomers are named pent-4-en-3-one and pent-3-en-2-one. Note that the carbonyl group cannot be bonded to either carbon atom of the C=C bond otherwise the carbon atoms would have five bonds, which is not possible.

Finally, we have the mono-unsaturated halogenoalkanes. When naming mono-unsaturated halogenoalkanes, the position of the C=C bond has priority over the position of the halogen atom and is assigned the lowest possible number. With two carbon atoms in the parent chain, a number is not required. **Figure 23** shows the structure of chloroethene, which has a chlorine atom bonded to one of the carbon atoms of the C=C bond.

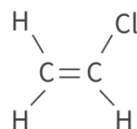


Figure 23. Chloroethene has two carbon atoms in the parent chain with a chlorine atom bonded to one of these atoms.

 More information for figure 23

The image is a diagram illustrating the chemical structure of chloroethene. It features two carbon atoms connected by a double bond, symbolized with an equal sign. One of the carbon atoms is bonded to a chlorine atom (Cl), and both carbon atoms have attached hydrogen atoms (H). Specifically, the carbon with the chlorine attached also has one hydrogen atom, while the other carbon has two hydrogen atoms. This structure reflects a mono-unsaturated halogenoalkane, emphasizing the priority of the carbon-carbon double bond (C=C) over the halogen positioning in nomenclature.

[Generated by AI]

With three carbon atoms in the parent chain there are three possible isomers. **Figure 24** shows the structures of the three isomers that have a bromine atom bonded to a different carbon atom in each isomer. The C=C bond takes priority over the position of the bromine atom and is assigned the lowest number accordingly. The isomers are named 1-bromopropene, 2-bromopropene and 3-bromopropene, respectively.

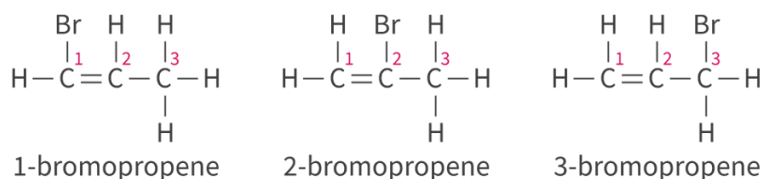


Figure 24. 1-bromopropene, 2-bromopropene and 3-bromopropene are structural isomers with the bromine atom bonded to a different carbon in each molecule.

 More information for figure 24.1

The image shows three structural isomers of bromopropene. Each structure has a three-carbon chain with variations in the position of a bromine atom.

- On the left, 1-bromopropene: The bromine atom (Br) is bonded to the first carbon. The first carbon has a double bond (C=C) with the second carbon, and a single bond with hydrogen (H).
- In the middle, 2-bromopropene: The bromine atom is bonded to the second carbon. The structure shows the first carbon double-bonded to the second carbon and single-bonded to hydrogen. The second carbon is also connected to the third carbon, which has another hydrogen.
- On the right, 3-bromopropene: The bromine atom is attached to the third carbon. The carbon chain features a double bond between the first and second carbons, with an H attached to each outermost carbon.

[Generated by AI]

With four carbon atoms in the parent chain the number of isomers increases. **Figure 25** shows the structures of four possible isomers with a chlorine atom bonded to different carbon atoms.

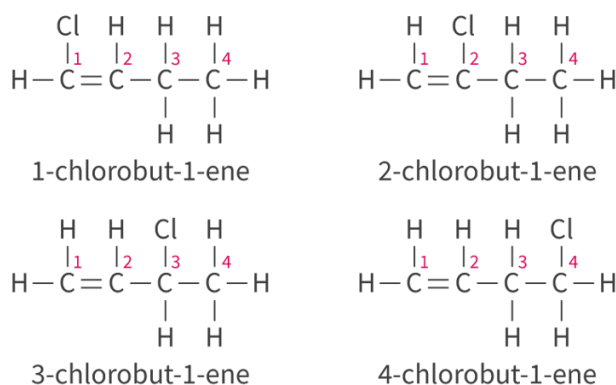


Figure 25. The four isomers of $\text{C}_4\text{H}_7\text{Cl}$ have the chlorine atom bonded to different carbon atoms in the parent chain.

 More information for figure 25

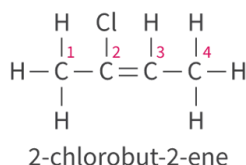
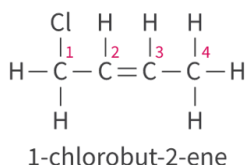
The image depicts the molecular structures of four isomers of $\text{C}_4\text{H}_7\text{Cl}$, labeled as 1-chlorobut-1-ene, 2-chlorobut-1-ene, 3-chlorobut-1-ene, and 4-chlorobut-1-ene. Each structure has a four-carbon parent chain, labeled with numbers 1 to 4, and a variable position of the chlorine (Cl) atom.

- 1. 1-chlorobut-1-ene:** The chlorine atom is bonded to the first carbon in the chain, with a double bond between the first and second carbons. The arrangement is $\text{H}-\text{C}(\text{Cl})=\text{C}-\text{C}-\text{H}$.
- 2. 2-chlorobut-1-ene:** The chlorine atom is attached to the second carbon, while the double bond remains between the first and second carbons, forming $\text{H}-\text{C}=\text{C}(\text{Cl})-\text{C}-\text{H}$.

3. **3-chlorobut-1-ene:** In this isomer, the chlorine is bonded to the third carbon, with the double bond still between the first and second carbons, arranged as $\text{H}-\text{C}=\text{C}-\text{C}(\text{Cl})-\text{H}$.
4. **4-chlorobut-1-ene:** Here, the chlorine attaches to the fourth carbon. The double bond is consistent between the first and second carbons, with the structure as $\text{H}-\text{C}=\text{C}-\text{C}-\text{C}(\text{Cl})-\text{H}$.

[Generated by AI]

There are another two isomers with the $\text{C}=\text{C}$ bond between carbon atom 2 and 3. Try to draw out the full structural formulas and name them yourself before checking your answers below.



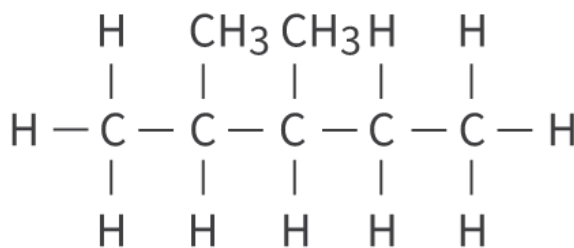
The two isomers are named 1-chlorobut-2-ene and 2-chlorobut-2-ene.

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 30 minutes
- **Activity type:** Individual activity

In this activity, you will practise naming organic compounds using IUPAC nomenclature rules.

Interactive 1 gives the full structural formula.


 Turn

Card 1 of 6



Interactive 1. Naming organic compounds from the full structural formula.

 More information for interactive 1

An interactive carousel selector features six pluck cards displaying full structural formulas of organic compounds, requiring users to identify the name of each structure.

Users can switch between the cards using the arrows present at the bottom right and bottom left. They can check the answer by selecting a turn button at the bottom of each card.

In the structures, carbon, oxygen, chlorine, hydrogen, and atoms and methyl and hydroxy groups are represented by their symbols.

Card 1:

Five C atoms are arranged in a straight line, connected by single bonds. The first C atom is single-bonded to H atoms on the top, bottom, and left. The second and third C atoms are each single-bonded to a CH₃ group on the top and an H atom at the bottom. The fourth C atom is single-bonded to H atoms at the top and bottom. The fifth C atom is single-bonded to H atoms on the top, bottom, and right.

Card 2:

Five C atoms are arranged in a straight line, connected by single and double bonds. The first C atom is double-bonded to the second C atom, while the remaining C atoms are connected by single bonds. The first C atom is single-bonded to H atoms at the top and left. The second C atom is single-bonded to one H atom at the top. The third and fifth C atoms are each single-bonded to H atoms at the top and bottom. The fifth C atom is also single-bonded to one H atom on the right. The fourth C atom is single-bonded to an OH group at the top and an H atom at the bottom.

Card 3:

Five C atoms are arranged in a straight line, connected by single and double bonds. The first C atom is double-bonded to the second C atom, while the remaining C atoms are connected by single bonds. The first C atom is single-bonded to H atoms at the top and left. The second C atom is single-bonded to one H atom at the top. The third and fourth C atoms are each single-bonded to H atoms at the top and bottom. The fifth C atom is double-bonded to an O atom at the top and single-bonded to an OH group at the bottom.

Card 4:

Five C atoms are arranged in a straight line, connected by single and double bonds. The third C atom is double-bonded to the fourth C atom, while the remaining C atoms are connected by single bonds. The first C atom is single-bonded to H atoms at the top, bottom, and left. The second C atom is single-bonded to H atoms at the top and bottom. The third and fourth C atoms are each single-bonded to one H atom at the top. The fifth C atom is double-bonded to an O atom at the top and single-bonded to an H atom at the bottom.

Card 5:

Six C atoms are arranged in a straight line, connected by single and double bonds. The third C atom is double-bonded to the fourth C atom, while the remaining C atoms are connected by single bonds. The first C atom is single-bonded to H atoms at the top, bottom, and left. The second C atom is single-bonded to H atoms at the top and bottom. The third and fourth C atoms are each single-bonded to one H atom at the top. The fifth C atom is double-bonded to an O atom at the top. The sixth C atom is single-bonded to H atoms at the top, bottom, and right.

Card 6:

Five C atoms are arranged in a straight line, connected by single and double bonds. The first C atom is double-bonded to the second C atom, while the remaining C atoms are connected by single bonds. The first C atom is single-bonded to H atoms at the top and left. The second C atom is single-bonded to one H atom at the top. The third and fourth C atoms are each single-bonded to H atoms at the top and bottom. The fifth C atom is single-bonded to a Cl atom at the top and H atoms at the right and bottom.

This interactivity helps users identify organic compounds, improve structural visualisation skills, and enhance understanding of chemical nomenclature.

Read below for the solution:

Card 1: 2,3-dimethylpentane

Card 2: Pent-4-en-2-ol

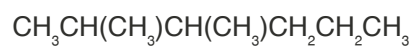
Card 3: Pent-4-enoic acid

Card 4: Pent-2-enal

Card 5: Hex-3-en-2-one

Card 6: 5-chloropent-1-ene

Interactive 2 gives the condensed structural formula.

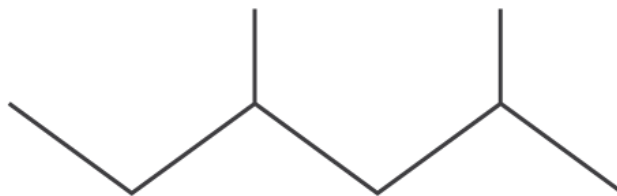
 Turn

Card 1 of 6



Interactive 2. Naming organic compounds from the condensed structural formula.

Interactive 3 gives the skeletal formula.

 Turn

Card 1 of 6



Interactive 3. Naming Organic Compounds from the Skeletal Formula.

 More information for interactive 3

An interactive carousel selector features six pluck cards displaying the skeletal structures of organic compounds, requiring users to identify the name of each structure.

Users can switch between the cards using arrows present in the bottom right and bottom left. They can check the answer by selecting a turn button at the bottom of each card.

In the structures, the zigzag line represents the carbon backbone, and hydrogen atoms are implied but not shown. Each endpoint and vertex of the zigzag represents a carbon atom.

Card 1:

A zigzag line represents a six-carbon chain. A short line branches off from each of the second and fourth carbon atoms from the right.

Card 2:

A zigzag line represents a five-carbon chain. A double bond is present between the second and third carbon atoms from the left. An OH group is attached to the first carbon atom.

Card 3:

A zigzag line represents a five-carbon chain. A double bond is present between the fourth and fifth carbon atoms from the left. An O atom is double-bonded to the first carbon atom, and an OH group is single-bonded to the same carbon atom.

Card 4:

A zigzag line represents a six-carbon chain. An O atom is double-bonded to the first carbon atom from the right. A double bond is present between the fifth and sixth carbon atoms.

Card 5:

A zigzag line represents a six-carbon chain. An O atom is double-bonded to the second carbon atom from the left. A double bond is present between the fifth and sixth carbon atoms.

Card 6:

A zigzag line represents a six-carbon chain. A double bond is present between the second and third carbon atoms from the left. A Br atom is bonded to the sixth carbon atom.

This interactivity helps users recognise and interpret skeletal structures of organic compounds.

Read below for the solution:

Card 1: 2,4-dimethylhexane

Card 2: Pent-2-ene-1-ol

Card 3: Pent-4-enoic acid

Card 4: Hex-5-enal

Card 5: Hex-5-en-2-one

Card 6: 6-bromo-hex-2-ene